

How I actually write

The pipeline behind "The Researcher's Three Outputs" — one researcher's own workflow, end to end, including which parts are paid and what to use instead.

Before anything: one routing question

ask it first, every time

Is the shape of this output already decided?

No — a new argument, a paper nobody has written yet → **Path A**, below.

Yes — a format you have produced before and will produce again → **Path B**, page 3.

Note what the question is **not**: it is not "is this important?". Two outputs of the same type can belong on different paths — and they need completely different machinery.

Path A — a new argument, from literature to manuscript

four steps

1. Collect the literature yourself

Google Scholar and your library. **Download the actual PDFs** — not summaries of them, not titles. The pipeline is only as honest as the papers you have really got.

Keep these private. Papers you downloaded under your institution's licence are yours to read, not yours to redistribute. Put them in your *own* notebook; never in a shared one.

2. Ground a notebook on them

Upload the set to **Gemini Notebook** and work through the four questions in order. It answers from your uploads only, and every claim carries a citation back to the passage — which is what makes the next step arguable rather than decorative.

```
#role a research assistant working only from my uploaded papers
#task 1) summarise what this literature agrees on 2) list what it disagrees about
      3) name the questions it leaves open 4) list the datasets these authors used
#constraints cite the source document for every point; where the papers do not say,
      write NOT IN SOURCES; invent nothing
#format four short sections, bullets
```

3. Spar in the chat panel — in your own language

The middle panel is a conversation, not a search box. This is where a literature review turns into **your** question: you argue, it argues back from the sources, and the gap it cannot fill from your uploads is usually where your contribution sits.

Think in the language you think in. Argue with the notebook in your first language, where your judgement is fastest and your scepticism sharpest. **Translate at the end, not at the start.** A question sharpened in English-as-a-second-language is often a blunter question.

```
#role a sceptical reviewer in my field
#task tell me which of these open questions is actually answerable with the data that exists
#constraints for each, name the dataset, the obstacle, and what would have to be true;
      say plainly which one you would abandon
#format ranked list, one paragraph each
```

4. Put everything in one folder, then open it as a project

Onto the desktop: the notebook's outputs, the source PDFs, your own notes, any data. **One folder.** Then **Claude desktop app** → **Claude Code** → **new project** → **name it** → **point it at that folder**. From here the assistant is no longer working from what you paste — it reads the whole folder, and it writes drafts back into it as files you keep.

The folder is the project. Not the chat. Chats end and are hard to find again; the folder is still there next year, and it is the same idea as the four-heading note you made on Monday — **your memory, on your machine, carried in and out.**

Why two different tools, and not one

a fair question

The notebook is for deciding what is true — it will not go beyond your uploads, and it cites the passage. **The project folder is for deciding what to say** — it reads everything at once and writes files back that you keep.

Reading tool, then making tool. Most trouble with AI writing comes from using one tool for both jobs: asking a grounded librarian to be a co-author, or asking a co-author to be careful about sources it was never given.



Structure first. Then one section at a time.

What happens inside the project folder — and the single habit that decides whether the draft is yours.

Never ask for the whole thing

the most expensive prompt in research

"Write the paper." What comes back is plausible, evenly written, and almost impossible to repair — because the errors are spread thinly through every section, and you meet all of them at once, at the end, when you are tired and the deadline is close. **You become the editor of a document whose shape you never chose.**

The fix is not a better prompt but a different order of work: **agree the structure first, then generate one section at a time.**

Step 5 — agree the structure

twenty minutes that save two days

With the folder open as a project, ask for the **shape**, not the prose. Three alternatives, so that choosing between them forces you to say what the paper actually claims.

```
#role a co-author who has read everything in this folder
#task propose 3 alternative structures for this paper
#constraints for each section give ONE line: the claim that section must establish,
    and which source or dataset in this folder establishes it;
    mark any section we cannot yet support with [NO EVIDENCE]; write no prose
#format three outlines, side by side
```

Then you edit it — by hand. Cut sections, reorder, rewrite the claim lines in your own words. This is the highest-value twenty minutes in the whole pipeline, and it is the part that is unmistakably authorship. **Save the agreed outline into the folder as a file;** every later step points back to it.

If two structures look equally good, you have not yet decided what the paper claims. That is a finding about your thinking, arriving early enough to be cheap.

Step 6 — generate one section, read it, save it, then the next

never two ahead

```
#role my co-author
#task draft ONLY section 3 of the agreed outline
#context the outline file, and the sections already written in this folder
#constraints use only sources in this folder; do not introduce claims that are not in
    the outline; write [CHECK] where you are unsure of a fact or a citation
#format continuous prose, ~600 words, no headings beyond the section title
```

- **Read it before anything else is generated.** An unread section becomes the foundation of the next one, and a wrong assumption compounds silently.
- **Fix it in your own words, then save it into the folder.** The corrected file — not the chat — is what the next section is built on.
- **Then, and only then, ask for the next section.**

Four things follow, and all four matter: errors stay **local** and cheap; each finished section becomes context, so the paper begins to **argue with itself**; the review burden spreads across the writing instead of landing in one block at the end; and you have **read every sentence as it appeared** — the practical meaning of being the author.

This is the same discipline that builds software. Plan first, then one file at a time, checked as you go — and it is why the rule generalises: **an agent's first deliverable should always be the plan, for you to approve.** Delegating a plan is a far larger delegation than delegating a draft.

When a section proves the outline wrong

it usually happens at section three

You will reach a section that cannot be written as agreed — thinner evidence than it looked, or a claim that belongs elsewhere. That is the process working, not failing.

Go back and change the outline file. Do not quietly patch the prose. Rewrite the claim line, note what changed and why, and continue. A draft that drifts away from its outline is how a paper ends up making two arguments and convincing nobody of either — and the referee will find the seam.

By the end the folder holds the source PDFs, the notebook's outputs, the dated outline, one file per section, your data. **That folder is the paper** — back it up like the manuscript it is.



The routine path, and what never changes

Path B is for the work you will do again next quarter — and the last box is the part no pipeline touches.

Path B — the output whose shape is already decided

hand over the pattern

Fill-in-the-blank handouts made from a slide deck. The quarterly policy brief in its same six sections. The organisation's newsletter. Nobody needs to invent the format — it was settled years ago, and inventing it again each time is the waste. For these, **Claude Cowork**: you hand over the whole task and receive the finished deliverable. What you hand over is not instructions but **the pattern**:

- **One finished example** you were happy with — last quarter's brief, the handout that worked.
- **The rule that made it good**, written out: section order, length, who reads it, what must never appear.
- **The new input**: this quarter's slides, this month's numbers.

Three signs a task belongs here: you have done it at least three times · you could show a finished example · **you can check the result in under five minutes**. If you cannot check it quickly, do not delegate it — you have not defined it well enough yet.

Both paths at a glance

and the free substitute for each

	Path A — new argument	Path B — routine output
Typical work	A paper nobody has written; a question you are still shaping	Quarterly brief, newsletter, handouts from slides
What you supply	Judgement, and the argument	The pattern, and the new input
Tool	Gemini Notebook (free) → Claude Code project (paid)	Claude Cowork (paid)
Free substitute	Stop after step 3 and use Claude.ai: paste the file in, copy the file out . Slower per session; identical thinking	Keep your example and rule in one note and paste all three parts into any free assistant

Plans as of **August 2026**. Pay only when the folder itself has become the bottleneck.

What the pipeline does not touch

the whole point

- **The research question is yours**. Ask for preconditions and objections, never for approval — it will encourage almost anything, fluently.
- **The contribution sentence is yours**, never generated. If you cannot write the one sentence saying what this adds, no tool here can rescue the draft.
- **Citations get the three checks**: does it exist · where was it published · who cites it.
- **The last edit is the author's**. Always.
- **Declare the assistance yourself**, in the method section. Detection is not proof — which is why you should not leave the account of your process to a detector.

The boundary that holds on both paths: unpublished data belonging to others, personal records and anything confidential do not go into a cloud tool. Not a rule about AI — the rule you already work under, arriving somewhere new.

The thirty-second version

if you remember nothing else

1. **Ask the routing question**. Is the shape already decided?
2. **Collect the real papers**, ground a notebook, and **argue with it in your own language** until the question is yours.
3. **One folder on the desktop**, opened as a project.
4. **Structure first** — three options, and you choose by hand.
5. **One section at a time**, read and saved before the next begins.
6. **Your last edit, your citations checked, your disclosure**.

Only steps **3** and **5** touch a paid tool, and each has a free substitute in the table above. **Everything that decides whether the paper is worth writing is in the other four**.

